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Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be differentiable. Suppose $\frac{\partial f}{\partial x}(x, y) = 0$ for all $(x, y) \in \mathbb{R}^2$. Then $f(x, y) = g(y)$ for some differentiable function $g : \mathbb{R} \rightarrow \mathbb{R}$.

Define $f_y : \mathbb{R} \rightarrow \mathbb{R}$ by $f_y(x) = f(x, y)$.

$$f'_y(x) = \lim_{x \rightarrow x_0; x \neq x_0} \frac{f_y(x) - f_y(x_0)}{x - x_0} = \lim_{x \rightarrow x_0; x \neq x_0} \frac{f(x, y) - f(x_0, y)}{x - x_0} = \frac{\partial f}{\partial x}(x_0, y) = 0,$$

so f_y is constant, say at c_y . Define $g : \mathbb{R} \rightarrow \mathbb{R}$ by $g(y) = f(0, y) = c_y$, so g is differentiable. Then for any $(x, y) \in \mathbb{R}^2$, $f(x, y) = f_y(x) = c_y = g(y)$, as desired.

Suppose $a \frac{\partial f}{\partial x} + b \frac{\partial f}{\partial y} = 0$ for $(a, b) \neq (0, 0)$. I claim that $f(x, y) = g(bx - ay)$ for some differentiable function g .

Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear transformation with matrix

$$\begin{bmatrix} a & -b \\ b & a \end{bmatrix}.$$

Since $(a, b) \neq 0$, we have $\det[T] = a^2 + b^2 \neq 0$, so T is invertible. Consider the composition $h : \mathbb{R}^2 \rightarrow \mathbb{R}$ given by $h = f \circ T$, which is differentiable. Note that $h'(u, v) = f'(x, y)T$. In particular,

$$\frac{\partial h}{\partial u}(u, v) = \begin{bmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = a \frac{\partial f}{\partial x} + b \frac{\partial f}{\partial y} = 0.$$

The previous result shows that $h(u, v) = g(v)$ for some differentiable $g : \mathbb{R} \rightarrow \mathbb{R}$. Hence, $f = h \circ T^{-1}$ has the form $f(x, y) = g\left(\frac{-bx + ay}{a^2 + b^2}\right)$, or up to scaling, $f(x, y) = g(bx - ay)$.

Indeed if f is of this form,

$$a \frac{\partial f}{\partial x} + b \frac{\partial f}{\partial y} = abg(bx - ay) - abg(bx - ay) = 0.$$

Assume $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ is twice differentiable with equality of mixed partials such that $\frac{\partial^2 f}{\partial x^2} - 2\frac{\partial^2 f}{\partial x \partial y} - 3\frac{\partial^2 f}{\partial y^2} = 0$. Solve for f .

Write $\frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} - 3\frac{\partial f}{\partial y} \right) + \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} - 3\frac{\partial f}{\partial y} \right) = 0$. Let $g : \mathbb{R}^2 \rightarrow \mathbb{R}$ be given by $g(x, y) = \frac{\partial f}{\partial x} - 3\frac{\partial f}{\partial y}$, so g is differentiable such that $\frac{\partial g}{\partial x} + \frac{\partial g}{\partial y} = 0$. The previous theory gives $g(x, y) = h(x - y)$ for some differentiable $h : \mathbb{R} \rightarrow \mathbb{R}$. Let $H : \mathbb{R} \rightarrow \mathbb{R}$ be an antiderivative of h (since h is continuous). Note that $f_0(x, y) = \frac{1}{4}H(x - y)$ satisfies

$$\frac{\partial f_0}{\partial x} - 3\frac{\partial f_0}{\partial y} = \frac{1}{4}h(x - y) + \frac{3}{4}h(x - y) = h(x - y).$$

Since scaled solutions of h work, take $f_0(x, y) = H(x - y)$ instead. Subtracting this particular solution, we have the PDE from before, so $f(x, y) = H(x - y) + K(3x + y)$ for twice differentiable functions H, K . Indeed,

$$\frac{\partial f}{\partial x} = h(x - y) + 3k(3x + y),$$

$$\frac{\partial f}{\partial y} = -h(x - y) + k(3x + y),$$

$$\frac{\partial^2 f}{\partial x^2} = h'(x - y) + 9k'(3x + y),$$

$$\frac{\partial^2 f}{\partial x \partial y} = -h'(x - y) + 3k'(3x + y),$$

$$\frac{\partial^2 f}{\partial y^2} = h'(x - y) + k'(3x + y).$$